MILITARY SPECIFICATION

SEMICONDUCTOR DEVICE, TRANSISTOR, PNP, GERMANIUM, LOW-POWER TYPE 2N331

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for an audio-frequency, PNP, germanium transistor.

1.2 Physical dimensions. See figure 1 (TO - 5).

1.3 Maximum ratings.*

P_C 1/ T_A = 25° C	V_{CBO}	V_{EBO}	V_{CEO}	I_C	T_{stg}
mW	Vdc	Vdc	Vdc	mAdc	° C
200	-30	-12	-16	200	-65 to +100

1/ Derate linearly at 2.67 mW/° C for $T_A > 25°$ C.

1.4 Primary electrical characteristics.

Limits	f_{hfb} V_{CB} = -6 Vdc I_E = 1 mAdc	h_{fe} V_{CE} = -6 Vdc I_C = 1.0 mAdc	h_{ib} V_{CB} = -6 Vdc I_E = 1.0 mAdc	C_{obo} V_{CB} = -6 Vdc I_E = 1 mAdc	NF V_{CB} = -6 Vdc I_E = 1.0 mAdc
	MHz		ohms	pf	db
Min	0.4	30	---	---	---
Max	---	70	50	50	20

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-S-19500 - Semiconductor devices, General Specification for.

* Wherever used herein, the frequency measurement unit "Hertz" (Hz) is equivalent to and replaces "cycles per second" (cps).

FSC 5961

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STANDARDS

MILITARY

MIL-STD-202 - Test Methods for Electronic and Electrical Component Parts.
MIL-STD-750 - Test Methods for Semiconductor Devices.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General. Requirements shall be in accordance with MIL-S-19500, and as specified herein.

3.2 Abbreviations, symbols, and definitions. The abbreviations, symbols, and definitions used herein are defined in MIL-S-19500.

3.3 Design, construction, and physical dimensions. - Transistors shall be of the design, construction, and physical dimensions shown on figure 1.

3.4 Performance characteristics. Performance characteristics shall be as specified in tables I and II.

3.5 Marking. The following marking specified in MIL-S-19500 may be omitted from the body of the transistor at the option of the manufacturer:

(a) Manufacturer's identification.
(b) Country of origin.

TABLE I. Group A inspection

Examination or test	MIL-STD-750		LTPD	Symbol	Limits		
	Method	Details			Min	Max	Unit
Subgroup 1			10				
Visual and mechanical examination	2071			---	---	---	---
Subgroup 2			5				
Collector-to-base cutoff current	3036	Bias cond. D; V_{CB} = -30 Vdc		I_{CBO}	---	-10	µAdc
Emitter-to-base cutoff current	3061	Bias cond. D; V_{EB} = -12 Vdc		I_{EBO}	---	-10	µAdc
Small-signal short-circuit forward-current transfer ratio	3206	V_{CE} = -6 Vdc; I_C = 1.0 mAdc		h_{fe}	30	70	---
Subgroup 3			5				
Small-signal short-circuit input impedance	3201	V_{CB} = -6 Vdc; I_E = 1.0 mAdc		h_{ib}	---	50	ohms
Small-signal short-circuit output admittance	3231	V_{CB} = -6 Vdc; I_E = 1.0 mAdc		h_{ob}	---	1	umhos
Open-circuit output capacitance	3236	V_{CB} = -6 Vdc; I_E = 1.0 mAdc; 100 kHz $\leq f \leq$ 1 MHz		Cobo	---	50	pf

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TABLE I. Group A inspection - Continued

Examination or test	MIL-STD-750		LTPD	Symbol	Limits		
	Method	Details			Min	Max	Unit
Subgroup 3 - Continued							
Noise figure	3246	V_{CB} = -6 Vdc; I_E = 1.0 mAdc; f = 1 kHz; R_G = 1,000 ohms; R_L = 20,000 ohms		NF	---	20	db
Subgroup 4			5				
Small-signal short-circuit forward-current transfer-ratio cutoff frequency	3301	V_{CB} = -6 Vdc; I_E = 1.0 mAdc		f_{hfb}	0.4	---	MHz
High-temperature operation:	---	T_A = +85° C		---	---	---	---
Collector-to-base cutoff current	3036	Bias cond. D; V_{CB} = -30 Vdc		I_{CBO}	---	-400	uAdc
Low-temperature operation:	---	T_A = -65° C		---	---	---	---
Small-signal short-circuit forward-current transfer ratio	3206	V_{CE} = -6 Vdc; I_C = 1.0 mAdc		h_{fe}	21	---	---

TABLE II. Group B inspection

Examination or test	MIL-STD-750		LTPD	Symbol	Limits		
	Method	Details			Min	Max	Unit
Subgroup 1			10				
Physical dimensions	2066	(See figure 1.)		---	---	---	---
Subgroup 2			15				
Solderability	2026	Omit aging		---	---	---	---
Thermal shock (temperature cycling)	1051	Test cond. B except in step 3, T = 100° C		---	---	---	---
Thermal shock (glass strain)	1056	Test cond. A		---	---	---	---
Seal (leak-rate)	---	Method 112 of MIL-STD-202, test cond. C procedure III; test cond. B for gross leaks		---	---	$5x10^{-7}$	atm cc/sec
Moisture resistance	1021			---	---	---	---
End points:							
Collector-to-base cutoff current	3036	Bias cond. D; V_{CB} = -30 Vdc		I_{CBO}	---	-10	μAdc
Small-signal short-circuit forward-current transfer ratio	3206	V_{CE} = -6 Vdc; I_C = 1.0 mAdc		h_{fe}	3[illegible]	70	---

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TABLE II. Group B inspection - Continued

Examination or test	MIL-STD-750		LTPD	Symbol	Limits		
	Method	Details			Min	Max	Unit
Subgroup 3			15				
Shock	2016	Nonoperating; 1,500 G, 0.5 msec, 5 blows in each orientation: X_1, Y_1, Y_2 and Z_1		---	---	---	---
Vibration fatigue	2046	Nonoperating		---	---	---	---
Vibration, variable frequency	2056			---	---	---	---
Constant acceleration	2006	20,000 G; in each orientation: X_1, Y_1, Y_2, and Z_1		---	---	---	---
End points:							
(Same as for subgroup 2.)							
Subgroup 4			15				
Terminal strength (lead fatigue)	2036	Test cond. E		---	---	---	---
Subgroup 5			15				
Salt atmosphere (corrosion)	1041			---	---	---	---
End points:							
(Same as for subgroup 2.)							
Subgroup 6			λ = 10				
High-temperature life (nonoperating)	1031	T_{stg} = +100° C		---	---	---	---
End points:							
Collector-to-base cutoff current	3036	Bias cond. D V_{CB} = -30 Vdc		I_{CBO}	---	-12	µAdc
Small-signal short-circuit forward-current transfer ratio	3206	V_{CE} = -6 Vdc I_C = 1.0 mAdc		h_{fe}	25	85	---
Subgroup 7			λ = 10				
Steady-state-operation life	1026	T_A = +25° ±3° C; P_C = 200 mW; V_{CB} = -25 Vdc		---	---	---	---
End points:							
(Same as for subgroup 6.)							

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4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection shall be in accordance with MIL-S-19500, and as specified herein.

4.2 Qualification inspection. Qualification inspection shall consist of the examinations and tests specified in tables I and II.

4.3 Quality conformance inspection. - Quality conformance inspection shall consist of groups A and B inspections.

4.4 Group A inspection. Group A inspection shall consist of the examinations and tests specified in table I.

4.5 Group B inspection. Group B inspection shall consist of the examinations and tests specified in table II.

4.6 Methods of examination and test. The methods of examination and test shall be as specified in tables I and II.

5. PREPARATION FOR DELIVERY

5.1 See MIL-S-19500, section 5.

6. NOTES

6.1 Notes. The notes specified in MIL-S-19500 are applicable to this specification.

6.2 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodians:
Army - EL
Navy - SH
Air Force - 11

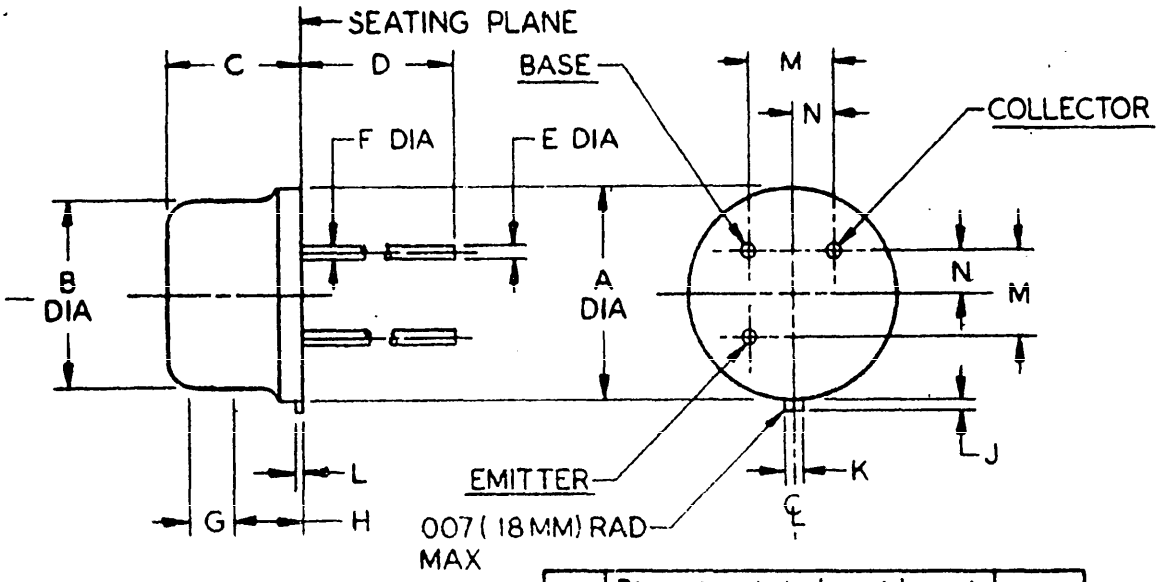
Review activities:
Army - EL, MU, MI
Navy - SH
Air Force - 11, 17, 85

User activities:
Army - SM
Navy - CG, MC, AS, OS
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Preparing activity:
Navy - SH

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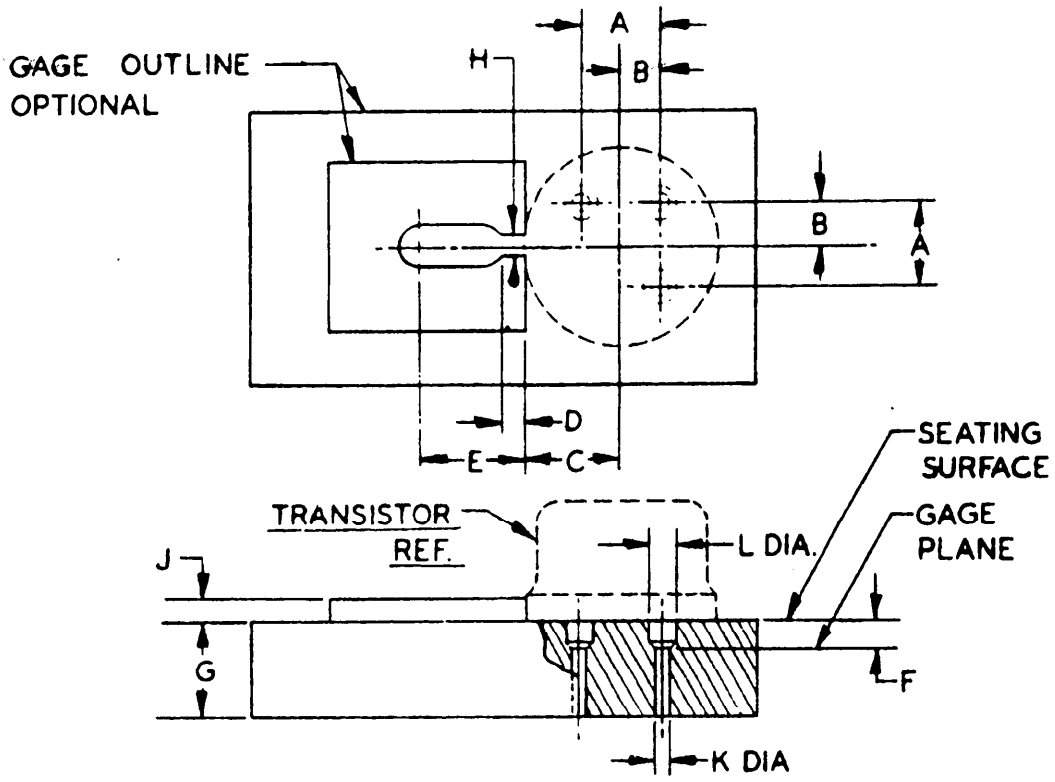
Ltr	Dimensions in inches with metric equivalents (mm) in parentheses (see note 1)		Notes
	Minimum	Maximum	
A	.335 (8.51)	.370 (9.40)	
B	.305 (7.75)	.335 (8.51)	
C	.240 (6.10)	.260 (6.60)	
D	1.500 (38.10)	1.750 (44.45)	9
E	.016 (.41)	.021 (.53)	2,9
F	.016 (.41)	.019 (.48)	3,9
G	.100 (2.54)		4
H			5
J	.029 (.74)	.045 (1.14)	8
K	.028 (.71)	.034 (.86)	
L	.009 (.23)	.125 (3.18)	
M	.1414 (3.59) nom		6
N	.0707 (1.80) nom		6

NOTES:

1. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
2. Measured in the zone beyond .250 (6.35 mm) from the seating plane.
3. Measured in the zone .050 (1.27 mm) and .250 (6.35 mm) from the seating plane.
4. Variations on Dim B in this zone shall not exceed .010 (.25 mm).
5. Outline in this zone is not controlled.
6. When measured in a gaging plane .054 +.001, -.000 (1.37 +.03, -.00 mm) below the seating plane of the transistor max dia leads shall be within .007 (.18 mm) of their true location relative to a maximum width tab. Smaller dia leads shall fall within the outline of the max dia lead tolerance. Figure 2 preferred measured method.
7. All leads electrically isolated from case.
8. Measured from the maximum diameter of the actual device.
9. All 3 leads.

FIGURE 1. Physical dimensions of transistor type 2N331 (TO-5).

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NOTES:
1. The following gaging procedure shall be used: The use of a pin straightener prior to insertion in the gage is permissible. The device being measured shall be inserted until its seating plane is .125±.010 (3.18±.25 mm) from the seating surface of the gage. A spacer may be used to obtain the .125 (3.18 mm) distance from the gage seat prior to force application. A force of 8 oz ±.5 oz shall then be applied parallel and symmetrical to the device's cylindrical axis. When examined visually after the force application (the force need not be removed) the seating plane of the device shall be seated against the gage.
2. The location of the tab locator, within the limits of dim C, will be determined by the tab and flange dimension of the device being checked.
3. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.

Ltr	Dimensions in inches with metric equivalents (mm) in parentheses (see note 3)	
	Minimum	Maximum
A	.1409 (3.58)	.1419 (3.60)
B	.0702 (1.78)	.0712 (1.81)
C	.182 (4.62)	.199 (5.05)
D	.009 (.23)	.011 (.28)
E	.125 (3.18) nom	
F	.054 (1.37)	.055 (1.40)
G	.372 (9.45)	.378 (9.60)
H	.0350 (.89)	.0355 (.90)
J	.150 (3.81) nom	
K	.0325 (.83)	.0335 (.85)
L	.0595 (1.51)	.0605 (1.54)

FIGURE 2. Gage for lead and tab location for transistor type 2N331.

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	☐ MANUFACTURER
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5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

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